

Tri Pham

p.tri@wustl.edu | (785) 331-6501 | linkedin.com/in/vothientripham | itstreefam.github.io

SUMMARY

- PhD candidate in Computer Science researching developer tools, code history interfaces, and interruption recovery, with publications in the Journal of Computer Languages and IEEE VL/HCC, and a paper under review at ACM CHI 2027.
- 5+ years of HCI research experience designing and running within-subjects user studies, and building code capture frameworks for VS Code and Chrome that record fine-grained programmer activity.
- Experienced educator and mentor: guest lecturing, teaching assistant for graduate and undergraduate courses, and advising master's thesis research.
- Technical proficiency in Python, JavaScript, C++, and extension/LLM APIs, with expertise in usability testing, think-aloud protocols, and user-centered design.

EDUCATION

Washington University in Saint Louis, McKelvey School of Engineering

Doctor of Philosophy in Computer Science (Advisor: Dr. Caitlin Kelleher)

Master of Science in Computer Science

Saint Louis, MO

Expected May 2027

May 2024

The University of Kansas (KU), School of Engineering

Bachelor of Science in Computer Science, Minor: Mathematics, Graduated with Honors

Lawrence, KS

May 2021

SKILLS

Instrumentation: Python, JavaScript, C++; VS Code and Chrome extension APIs; OpenAI and Claude APIs

Techniques: Usability Testing, Think-Aloud Protocols, Contextual Inquiry, Interviews and Surveys, Task Analysis, User-Centered and Iterative Design, Prototyping, Heuristic Evaluation, Cognitive Walkthrough, A/B Testing, Log Data Analysis

Software: VS Code, PyCharm, Git/GitHub, Canvas, Piazza, Microsoft Office, Google Workspace, Zoom, Slack

Languages: Vietnamese (Native) and English (Fluent)

PROJECTS

Rapid Prototyping

Fall 2023 – Present

- Built different applications and polished their tasks to be used as materials for user studies. Applications ranged from common algorithmic focus like Sorting Visualizer and Pathfinding Visualizer to more game focus like Doodle Jump and Tile Matching Puzzle.

Natural Language Processing

Spring 2024

- Fine-tuned a Flan-T5-XL model from the paper *InstructBLIP: Towards General-purpose Vision-Language Models with Instruction Tuning* by broadening instruction templates to improve model response to complex natural language tasks and optimizing computational efficiency via Low-Rank Adaptation (LoRA) method.

RESEARCH EXPERIENCE

Graduate Research Assistant, Dr. Caitlin Kelleher's Lab

September 2022 – Present

Washington University in Saint Louis, Department of Computer Science & Engineering | Saint Louis, MO

- **Developed and maintained an integrated code capture framework** for VS Code and Chrome that records fine-grained development activity (code edits, executions, web resources) to enable detailed analysis of programmer behavior and code evolution.
- Designed and conducted 28 within-subjects user study sessions comparing chronological and sub-goal-based history interfaces for context recovery, and **formalized recovery as a multi-stage process rather than a single event** (VL/HCC 2025).
- Led a within-subjects study with 16 developers resuming programming tasks after 10-minute and one-week interruptions (**first systematic comparison between short-term and long-term interruptions, exploring developers' change in history and information needs**); found time to first edit doubled after a week (2.5 to 5 minutes), driven by locating work rather than re-reading code, and that preferred record type reversed with interruption length (CHI 2027, under review).
- **Exploring LLM-supported interfaces that answer questions about codebase history** (why code changed, how it implements a behavior) using captured development context.
- Mentoring two master's students on developer tools research (Fall 2026 – Present); previously mentored a student from undergraduate senior through master's degree, advising thesis projects on developer tools and interruption recovery (Fall 2024 – Summer 2025).

Research Intern, Dr. Gerald Schweiger's Lab

Summer 2020

Graz University of Technology (TU Graz) | Graz, Austria

- Synthesized state-of-the-art findings in building energy simulation with peer researchers, and helped develop and validate methods to forecast building energy usage.
- Researched feature selection and prediction models for building energy consumption; tuned hyperparameters and built visualizations for statistical models.

PUBLICATIONS & PRESENTATIONS

- **Vo Thien Tri Pham**, Rose Liu, Haixin Zhou, and Caitlin Kelleher. Exploring History and Information Needs for Extended Interruptions. Submitted to the *ACM CHI Conference on Human Factors in Computing Systems* (CHI 2027). Under review.
- **Vo Thien Tri Pham** and Caitlin Kelleher. 2025. Code histories: Documenting development by recording code influences and changes in code. *Journal of Computer Languages*, 82, Article 101313. <https://doi.org/10.1016/j.cola.2024.101313>
- **Vo Thien Tri Pham**, Haixin Zhou, and Caitlin Kelleher, "Interruptions and Recovery: Leveraging Dynamic Code History in Development," *2025 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, Raleigh, NC, USA, 2025, pp. 7-18, doi: 10.1109/VL-HCC65237.2025.00012.
- **Presenter**, IEEE VL/HCC 2025, North Carolina State University, Raleigh, NC (October 2025)

TEACHING EXPERIENCE

Teaching Assistant, Washington University in St. Louis

August 2025 – Present

CSE 5104 Data Mining (Fall 2026); CSE 2506 Intro to Human-Centered Design (Fall 2026, Fall 2025)

- Delivered a lecture on Cognitive Load Theory and its applications in interface design (Fall 2025).
- Grade homework, exams, and programming assignments covering data preprocessing, modeling, and data analysis; help develop solution keys and grading rubrics.
- Hold weekly office hours, answer student questions on Piazza, and maintain Canvas gradebooks while following course policies on fair treatment and student data privacy.

Supplemental Instructor, KU School of Engineering (Supervisor: Dr. David Johnson)

August 2020 – May 2021

- Supported 50 students per semester in machine learning coursework (supervised/unsupervised learning, neural networks, model evaluation), including one-on-one Python debugging help.
- Prepared targeted review materials from lecture content and exam patterns, and reported recurring misconceptions to the instructor to inform instructional adjustments.

LEADERSHIP & CAMPUS INVOLVEMENT

Resident Assistant, KU Jayhawker Towers

April 2020 – May 2020

- Stepped into the role during COVID-19 to support students remaining on campus; handled on-call response and enforced CDC and state health protocols.

Food Board Manager, KU Pearson Scholarship Hall

August 2019 – April 2020

- Managed a \$100,000 annual food budget; remaining funds became relief support for students affected by COVID-19.

Public Service Intern, KU Honors Program

May 2018 – May 2019

Orientation Assistant, KU First-Year Experience

March 2018 – April 2019

SCHOLARSHIPS & AWARDS

- KU International Excellence Award (Undergraduate Full Tuition Scholarship) 2017 – 2021
- KU Honors Opportunity Award 2020
- Nelson Scholarship (2019–2020); Gerlach Scholarship, Garmin Excellence Scholarship, Garmin EECS Scholarship 2018 – 2020
- 2nd Place, 36th Annual Mathematics Prize Competition, KU Department of Mathematics 2018